# Supplementary Material

Multi-objective Pareto Hyperparameter Optimization for Joint Detection and Localization of Forged Identity Documents

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This document is a companion to the Pattern Recognition Letters submission *Multi-objective Pareto Hyperparameter Optimization for Joint Detection and Localization of Forged Identity Documents* (DocVerify). It collects the complete experimental tables that, owing to the journal's strict page limit, are summarized in only one or two sentences in the main paper or in the response to reviewers. All numbers are computed from the released result CSVs (archived together with this document); section references (e.g. Sec. 4.3) point to the main paper. Unless stated otherwise, all blind-test results use the held-out 15% FantasyID partition, $224 \times 224$ inputs, and the Pareto-selected configuration of Table 2 in the main paper. Statistical tests are paired two-sided Wilcoxon signed-rank tests with Holm correction across metrics within an experiment; effect sizes are paired Cohen's $d$.

## Contents



## S1 Data-augmentation ablation

Sec. 4.1 states that standard augmentation degraded every metric on the FantasyID holdout. Table S1 gives the full 30-seed comparison. The augmented model applies, in the training loader only, random rotation ($\pm 10°$), brightness, contrast, Gaussian blur, and JPEG compression, using the *same* Pareto configuration. On the clean synthetic holdout, augmentation significantly degrades every metric (all Holm $p < 10^{-7}$) and inflates variance by 2–5×, confirming the paper's no-augmentation choice for this dataset.

## S2 Class-imbalance (pos_weight) sensitivity

Sec. 3.1 notes that the mild class imbalance (2.5:1, attack-majority) leaves PR-AUC robust and that a `pos_weight` sweep confirms no significant effect. Table S2 reports the three settings (inverse-frequency 0.397, neutral 1.0, and the attack-favouring 2.518), each over 30 seeds. No metric differs significantly from the neutral baseline after Holm correction (all $p_{\text{Holm}} > 0.5$), so no loss re-weighting is applied in the main model.

**Table S1:** Augmentation ablation (blind test, 30 seeds). $\Delta$: aug − no_aug; $d$: paired Cohen's $d$; $p_{\text{Holm}}$: Holm-corrected paired Wilcoxon. All differences significant.

| Metric | No aug (ours) | Aug | $\Delta$ | $d$ | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|
| PR-AUC | $0.9967 \pm 0.0021$ | $0.9577 \pm 0.0457$ | $-0.0390$ | $-0.84$ | $1.5 \times 10^{-8}$ |
| Dice | $0.8750 \pm 0.0180$ | $0.7797 \pm 0.0396$ | $-0.0953$ | $-2.10$ | $9.3 \times 10^{-9}$ |
| BAcc | $0.9573 \pm 0.0184$ | $0.8302 \pm 0.1007$ | $-0.1271$ | $-1.20$ | $5.6 \times 10^{-9}$ |
| $\text{F1}_1$ | $0.9651 \pm 0.0152$ | $0.8593 \pm 0.1061$ | $-0.1058$ | $-0.97$ | $1.1 \times 10^{-8}$ |
| $\text{F1}_{\text{macro}}$ | $0.9423 \pm 0.0234$ | $0.7998 \pm 0.1136$ | $-0.1425$ | $-1.19$ | $7.5 \times 10^{-9}$ |

**Table S2:** `pos_weight` sensitivity (blind test, 30 seeds). Baseline is `pos_weight`$= 1.0$. No comparison against the baseline reaches significance after Holm correction.

| `pos_weight` | PR-AUC | Dice | BAcc | Rec. attack | Rec. bonafide |
|---|---|---|---|---|---|
| 0.397 (inv. freq.) | $0.9973 \pm 0.0018$ | $0.8793 \pm 0.0168$ | $0.9614 \pm 0.0146$ | 0.9431 | 0.9797 |
| 1.0 (baseline) | $0.9967 \pm 0.0021$ | $0.8750 \pm 0.0180$ | $0.9573 \pm 0.0184$ | 0.9435 | 0.9712 |
| 2.518 (attack-favouring) | $0.9967 \pm 0.0019$ | $0.8688 \pm 0.0240$ | $0.9552 \pm 0.0164$ | 0.9435 | 0.9669 |

## S3 Selection criterion: Pareto vs. scalar (full results)

Table 3 of the main paper reports a compact form of the selection-criterion study. Table S3 gives the complete 30-seed results with 95% confidence intervals for all five metrics, and Table S4 the paired statistics. The two reported objectives quantify the two tasks: PR-AUC (area under the precision–recall curve) is a threshold-independent measure of the detection head, summarizing precision over the full range of recall and remaining informative under class imbalance; Dice (the Sørensen–Dice overlap coefficient, $2|A \cap B|/(|A| + |B|)$ between the predicted and ground-truth forged-region masks) measures localization quality. Our Pareto criterion selects the trial closest to the ideal point $(1, 1)$ in the normalized PR-AUC/Dice plane, whereas `by_prauc` and `by_dice` take the global argmax of validation PR-AUC or validation Dice, respectively.

Significance is assessed with paired two-sided Wilcoxon signed-rank tests over the 30 shared seeds, with Holm correction across the five metrics within each comparison; effect sizes are paired Cohen's $d$. Pareto improves localization over `by_prauc` (Dice $\Delta = +0.0161$, $d = 0.59$, $p_{\text{Holm}} = 0.043$) while not degrading detection ($\Delta$PR-AUC $= +0.0014$, $p_{\text{Holm}} = 0.062$, n.s.), and improves *both* objectives over `by_dice` (PR-AUC $\Delta = +0.0019$, $d = 0.64$, $p_{\text{Holm}} = 0.0072$; Dice $\Delta = +0.0185$, $d = 1.00$, $p_{\text{Holm}} = 0.00039$), the latter a large effect. All gains over `by_dice` except $\text{F1}_1$ reach significance after correction (Table S4).

**Table S3:** Selection-criterion comparison (blind test, 30 seeds). Mean ± Std with 95% CI in brackets.

| Metric | **Pareto (ours)** | `by_prauc` | `by_dice` |
|---|---|---|---|
| PR-AUC | $0.9967 \pm 0.0021$ | $0.9953 \pm 0.0021$ | $0.9948 \pm 0.0027$ |
| | [0.9959, 0.9975] | [0.9945, 0.9961] | [0.9938, 0.9958] |
| Dice | $0.8750 \pm 0.0180$ | $0.8589 \pm 0.0230$ | $0.8566 \pm 0.0147$ |
| | [0.8683, 0.8818] | [0.8504, 0.8675] | [0.8511, 0.8620] |
| BAcc | $0.9573 \pm 0.0184$ | $0.9481 \pm 0.0146$ | $0.9463 \pm 0.0170$ |
| | [0.9504, 0.9642] | [0.9427, 0.9536] | [0.9400, 0.9527] |
| $\text{F1}_1$ | $0.9651 \pm 0.0152$ | $0.9569 \pm 0.0122$ | $0.9564 \pm 0.0121$ |
| | [0.9594, 0.9708] | [0.9523, 0.9614] | [0.9519, 0.9609] |
| $\text{F1}_{\text{macro}}$ | $0.9423 \pm 0.0234$ | $0.9294 \pm 0.0184$ | $0.9283 \pm 0.0184$ |
| | [0.9335, 0.9510] | [0.9225, 0.9362] | [0.9214, 0.9352] |

**Table S4:** Paired statistics, Pareto vs. each scalar criterion (30 seeds). $\Delta$: Pareto $-$ scalar; $d$: paired Cohen's $d$; $p_{\mathrm{Holm}}$: Holm-corrected paired Wilcoxon; ✓: significant at $p_{\mathrm{Holm}} < 0.05$.

| Comparison | Metric | $\Delta$ | $d$ | $p_{\mathrm{Holm}}$ | sig. |
|---|---|---|---|---|---|
| Pareto vs `by_prauc` | PR-AUC | +0.0014 | 0.49 | 0.062 | |
| | Dice | +0.0161 | 0.59 | 0.043 | ✓ |
| | BAcc | +0.0092 | 0.40 | 0.061 | |
| | $F1_1$ | +0.0082 | 0.45 | 0.066 | |
| | $F1_{\mathrm{macro}}$ | +0.0129 | 0.46 | 0.050 | ✓ |
| Pareto vs `by_dice` | PR-AUC | +0.0019 | 0.64 | 0.0072 | ✓ |
| | Dice | +0.0185 | 1.00 | 0.00039 | ✓ |
| | BAcc | +0.0110 | 0.45 | 0.040 | ✓ |
| | $F1_1$ | +0.0087 | 0.44 | 0.066 | |
| | $F1_{\mathrm{macro}}$ | +0.0140 | 0.47 | 0.043 | ✓ |

## S4 Architecture-agnostic validation (ResNet-18)

Sec. 4.6 states that the identical MOTPE protocol, applied to an ImageNet-pretrained ResNet-18 backbone, yields comparable blind-test performance and that neither model dominates. This section gives the full experimental detail.

**Protocol (identical to the main paper).** We run a single MOTPE study with exactly the search space of Table 1 (learning rate, weight decay, dropout, decoder width $C_{\mathrm{dec}}$, and $\lambda_{\mathrm{mask}}$), the same 5 inner folds, the same 50 trials, the same 15 epochs per trial, the same MOTPE sampler, and the same Pareto selection by minimum Euclidean distance to the ideal point $(1, 1)$. To bound cost we run one study (the outer-fold-0 analogue) rather than the full 10-fold nested loop. The Pareto-selected configuration is then retrained on the same blind holdout over 30 seeds, with early stopping (patience 12 on the validation distance to the ideal point) and the classification threshold fixed from the development validation split, never from the holdout, exactly as for DocVerify. The study is in-memory and writes no weights; it only reads DocVerify's released blind-test CSV as the paired baseline, so it leaves every paper result untouched.

**Architecture integration.** Only the encoder changes: the six-block Patel CNN is replaced by a torchvision `resnet18` initialized with the `ResNet18_Weights.IMAGENET1K_V1` checkpoint, downloaded automatically from `https://download.pytorch.org/models/resnet18-f37072fd.pth`. ImageNet normalization (mean $[0.485, 0.456, 0.406]$, std $[0.229, 0.224, 0.225]$) is applied inside the forward pass (inputs arrive in $[0, 1]$ as in the main pipeline). The five encoder taps are the stem ($112 \times 112$, 64 ch), `layer1` ($56 \times 56$, 64 ch), `layer2` ($28 \times 28$, 128 ch), `layer3` ($14 \times 14$, 256 ch), and `layer4` ($7 \times 7$, 512 ch, bottleneck). The 512-d global-average-pooled bottleneck feeds the *same* four-layer classification head ($512 \to 32 \to 16 \to 16 \to 1$, dropout and LeakyReLU as in the paper). For localization, a $1 \times 1$ convolution projects the bottleneck to $C_{\mathrm{dec}}$, and four U-Net decoder blocks ($7 \to 14 \to 28 \to 56 \to 112$) concatenate the `layer3`, `layer2`, `layer1`, and stem skips in turn; a final bilinear upsampling to $224 \times 224$ and a $1 \times 1$ convolution produce the mask logit. The decoder and both heads are identical to DocVerify; only the feature extractor differs.

Table S5 lists the complete configuration applied to both studies: the fixed training settings, the MOTPE protocol, the blind-test retraining, and the search space with the Pareto-selected value for each backbone. The two studies share everything except the encoder and the resulting selected values.

**Table S5:** Complete hyperparameter configuration for the DocVerify and ResNet-18 MOTPE studies. All fixed settings and the optimization protocol are identical; only the encoder and the Pareto-selected values differ. Stratified group $k$-fold and MOTPE sampler are both seeded at 42.

| **Fixed training settings (both backbones)** | | |
|---|---|---|
| Input | $224 \times 224$, normalized to $[0, 1]$ | |
| Batch size | 64 | |
| Optimizer | AdamW | |
| LR scheduler | ReduceLROnPlateau (factor 0.5, patience 3) | |
| Mixed precision | AMP (FP16/FP32) | |
| Gradient clip | 1.0 (L2 norm) | |
| Activation | LeakyReLU ($\alpha = 0.2$, fixed) | |
| Mask threshold | 0.5 | |
| Loss | $\mathrm{BCE_{cls}} + \lambda_{\mathrm{mask}}\,(\mathrm{BCE} + \mathrm{Dice})_{\mathrm{seg}}$ | |
| **MOTPE protocol (both backbones)** | | |
| Inner folds | 5 (stratified group $k$-fold) | |
| Trials | 50 (MOTPE sampler) | |
| Epochs / trial | 15 (validate every 2, patience 1) | |
| Selection | min. distance to ideal point $(1, 1)$ | |
| **Blind-test retraining (both backbones)** | | |
| Epochs | up to 100, early stopping patience 12 | |
| Seeds | 30 (seeds 42–71) | |
| Threshold | from dev. validation split (never holdout) | |
| **Search space and Pareto-selected value** | | |
| Parameter | Range / set | DocVerify / ResNet-18 |
| $\eta$ | $[5\times10^{-5},\ 9\times10^{-4}]$ log | $5.21\times10^{-4}$ / $3.68\times10^{-4}$ |
| $\lambda_{\mathrm{wd}}$ | $[10^{-7},\ 10^{-4}]$ log | $4.27\times10^{-5}$ / $1.45\times10^{-6}$ |
| $p$ (dropout) | $[0.1,\ 0.4]$ uniform | 0.103 / 0.381 |
| $C_{\mathrm{dec}}$ | $\{96, 128, 192, 256\}$ | 192 / 256 |
| $\lambda_{\mathrm{mask}}$ | $[0.5,\ 3.0]$ uniform | 2.46 / 2.69 |
| Distance to ideal | 0.316 (DocVerify) / 0.164 (ResNet-18) | |

**Result.** Table S6 gives the full 30-seed comparison. The ResNet-18 Pareto-selected configuration (distance to ideal = 0.164) is $\eta = 3.68 \times 10^{-4}$, $\lambda_{\mathrm{wd}} = 1.45 \times 10^{-6}$, dropout = 0.381, $C_{\mathrm{dec}} = 256$, $\lambda_{\mathrm{mask}} = 2.69$, the last consistent with DocVerify's 2.46, confirming that the Pareto balance the method discovers is stable across backbones. DocVerify is better on PR-AUC; ResNet-18 is better on Dice and the F1 metrics; balanced accuracy does not differ significantly. Neither model dominates, so the optimization methodology, not a specific encoder, is what generalizes.

# S5 Cross-domain generalization (SIDTD)

Sec. 4.6 reports that the FantasyID-trained model is near chance zero-shot on the out-of-domain SIDTD dataset, but recovers strong performance when the same architecture is retrained on the SIDTD training partition. This section details the full adaptation curve.

**Target domain.** SIDTD (Boned et al., 2024) is a synthetic dataset of ID and travel documents built from real document templates, a distinct visual domain from FantasyID's invented *fantasy*

**Table S6:** DocVerify vs. ResNet-18 under the identical MOTPE protocol (blind test, 30 seeds). $\Delta$: DocVerify − ResNet-18; $d$: paired Cohen's $d$; $p_{\text{Holm}}$: Holm-corrected paired Wilcoxon.

| Metric | DocVerify | ResNet18-MOTPE | $\Delta$ | $d$ | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|
| PR-AUC | $0.9967 \pm 0.0021$ | $0.9942 \pm 0.0025$ | $+0.0024$ | $+0.75$ | $2.8 \times 10^{-4}$ |
| Dice | $0.8750 \pm 0.0180$ | $0.8894 \pm 0.0111$ | $-0.0144$ | $-0.60$ | $8.1 \times 10^{-3}$ |
| BAcc | $0.9573 \pm 0.0184$ | $0.9629 \pm 0.0108$ | $-0.0055$ | $-0.21$ | 0.285 (n.s.) |
| $F1_1$ | $0.9651 \pm 0.0152$ | $0.9779 \pm 0.0069$ | $-0.0128$ | $-0.67$ | $3.8 \times 10^{-3}$ |
| $F1_{\text{macro}}$ | $0.9423 \pm 0.0234$ | $0.9613 \pm 0.0114$ | $-0.0190$ | $-0.63$ | $4.0 \times 10^{-3}$ |

identity cards. We use only the *templates* subset present on disk (2,222 images: 1,000 bonafide and 1,222 attack), not the full SIDTD video collection. Masks are derived from the provided annotations at the same $224 \times 224$ resolution as the main pipeline. The dataset is obtained from its official distribution at http://datasets.cvc.uab.es/SIDTD/ (doi:10.1038/s41597-024-04160-9).

**Zero-shot.** The 30 FantasyID blind-test multitask checkpoints are evaluated directly on the SIDTD templates with no adaptation, at a fixed classification threshold of 0.5. We headline PR-AUC and ROC-AUC because they are threshold-independent and therefore measure whether the model still *ranks* bonafide vs. attack across domains; balanced accuracy and F1 at a fixed 0.5 threshold are unreliable cross-domain and are reported only for completeness.

**Few-shot and full retraining.** For each budget $n_{\text{shots}} \in \{10, 25, 50, 100, 200, 400\}$ we draw a class-stratified sample from the SIDTD training partition and fine-tune the same architecture, repeating over 25 independent sample draws (seeds) per budget to capture sampling variance; the full-retraining point ($n_{\text{shots}} = 1777$) uses an 80/20 stratified split of the templates subset (1,777 train, 445 test). Table S7 gives the resulting curve from zero-shot through few-shot to full retraining. Few-shot adaptation up to 400 samples stays near chance; only full in-domain retraining recovers strong detection (PR-AUC 0.878, ROC-AUC 0.826), confirming that the gap reflects domain shift rather than architectural capacity. Section S6 further shows that training-time augmentation does not substitute for in-domain data.

**Table S7:** SIDTD adaptation curve. $n_{\text{shots}} = 0$ is zero-shot (30 seeds); $n_{\text{shots}} \geq 10$ are few-shot/full retrainings (25 seeds each). Templates subset (1 000 bonafide, 1 222 attack).

| $n_{\text{shots}}$ | PR-AUC | ROC-AUC | BAcc | $F1_{\text{macro}}$ |
|---|---|---|---|---|
| 0 (zero-shot) | $0.5554 \pm 0.0110$ | $0.5041 \pm 0.0094$ | 0.5010 | — |
| 10 | $0.5485 \pm 0.0084$ | $0.4893 \pm 0.0147$ | $0.5002 \pm 0.0135$ | $0.4841 \pm 0.0199$ |
| 25 | $0.5595 \pm 0.0089$ | $0.4996 \pm 0.0092$ | $0.5046 \pm 0.0146$ | $0.4755 \pm 0.0255$ |
| 50 | $0.5575 \pm 0.0102$ | $0.4960 \pm 0.0128$ | $0.4953 \pm 0.0167$ | $0.4652 \pm 0.0207$ |
| 100 | $0.5706 \pm 0.0208$ | $0.5060 \pm 0.0199$ | $0.5032 \pm 0.0203$ | $0.4848 \pm 0.0259$ |
| 200 | $0.5774 \pm 0.0170$ | $0.5089 \pm 0.0144$ | $0.5031 \pm 0.0182$ | $0.4853 \pm 0.0289$ |
| 400 | $0.5912 \pm 0.0238$ | $0.5123 \pm 0.0171$ | $0.5028 \pm 0.0184$ | $0.4843 \pm 0.0349$ |
| **1 777 (full)** | $\mathbf{0.8780 \pm 0.0161}$ | $\mathbf{0.8263 \pm 0.0228}$ | $\mathbf{0.7188 \pm 0.0349}$ | $\mathbf{0.7142 \pm 0.0333}$ |

## S6 Augmentation does not close the cross-domain gap

As a follow-up to Sections S1 and S5, we tested whether augmentation-trained checkpoints generalize better zero-shot to SIDTD. Table S8 shows a null result: both the augmented and non-augmented models remain at chance, with no significant difference in detection quality after Holm correction. Augmentation therefore does not substitute for in-domain data; closing the gap requires multi-source training (future work, Sec. 6).

**Table S8:** Zero-shot SIDTD generalization, augmented vs. non-augmented FantasyID training (30 seeds each). $\Delta$: aug − no_aug; $p_{\mathrm{Holm}}$: Holm-corrected paired Wilcoxon.

| Metric | No aug | Aug | $\Delta$ | $p_{\mathrm{Holm}}$ |
|---|---|---|---|---|
| PR-AUC | $0.5554 \pm 0.0110$ | $0.5639 \pm 0.0096$ | +0.0085 | 0.077 (n.s.) |
| ROC-AUC | $0.5041 \pm 0.0094$ | $0.5089 \pm 0.0060$ | +0.0048 | 0.199 (n.s.) |

**Data availability.** The result CSVs underlying every table above are released in the reproducibility archive accompanying this document. The full source code and experiment scripts are at `https://github.com/HernanDiaz/IDVerify`. Datasets and pretrained weights are obtained from their original sources: FantasyID from `https://zenodo.org/records/17063366`; SIDTD from `http://datasets.cvc.uab.es/SIDTD/` (doi:10.1038/s41597-024-04160-9); and the ResNet-18 ImageNet weights from torchvision (`ResNet18_Weights.IMAGENET1K_V1`).